Problem – Solution Fit

Date	5 January 2026
Team ID	LTVIP2026TMIDS47882
Project Name	Heart Disease Analysis
Maximum Marks	2 Marks

Problem – Solution Fit

The Problem–Solution Fit phase ensures that the identified healthcare challenges are effectively addressed by the proposed solution. In this project, the major challenge was analysing large-scale heart disease datasets involving patient demographics, medical history, lifestyle choices, and clinical indicators. To overcome this, an interactive Tableau dashboard was designed to provide clear visual insights, dynamic filtering options, and storytelling dashboards. The solution directly aligns with user needs by simplifying data interpretation, highlighting risk factors, and improving preventive decision-making efficiency.

Problem Area	Identified Problem	Proposed Solution	How the Solution Fits the Problem
Data Interpretation	Heart disease data is stored in raw formats and difficult to interpret.	Developed interactive Tableau dashboard with visual charts and graphs.	Converts raw health data into clear visual insights for easy understanding.
Risk Factor Comparison	Difficult to compare patient groups across demographics and lifestyle indicators.	Created dashboards with filters for age, gender, BMI, cholesterol, and smoking habits.	Enables quick and clear comparison of risk factors across patient groups.
Regional Analysis	No clear visibility of heart disease prevalence across regions.	Designed heatmaps and pie charts for regional distribution.	Helps identify which regions show higher prevalence and trends.

High-Risk Segmentation	Hard to identify highrisk and low-risk patient groups.	Implemented segmentation dashboards to classify patients by risk levels.	Allows dynamic identification of vulnerable groups for targeted interventions.
Trend Identification	Lifestyle and clinical trends are unclear and time-consuming to analyze manually.	Created trend line charts to visualize correlations and seasonal variations.	Clearly displays lifestyle impacts and clinical fluctuations over time.
Decision Support	Manual analysis slows preventive policy and awareness campaign planning.	Added interactive filters (Age, Gender, Region, Lifestyle) and storytelling dashboards.	Improves decisionmaking speed and empowers healthcare providers and policymakers with actionable data.

1. CUSTOMER SEGMENT(S) — C5

- Healthcare providers, cardiologists, government health departments, public health researchers, data analysts, policy makers.

6. CUSTOMER CONSTRAINTS — CC

- Heart disease data is stored in large, raw formats making interpretation difficult.
- Manually comparing patient demographics, lifestyle factors and clinical indicators is time-consuming.

5. AVAILABLE SOLUTIONS — A5

- Currently, the data is in large raw formats. making interpretation difficult. Manually comparing patient demographics, lifestyle factors, and clinical indicators is time-consuming.

3. JOBS-TO-BE-DONE / PROBLEMS — J&P

- Interpreting raw health data is challenging,
- Comparing risk factors across regions and demographics is difficult.
- Identifying high risk groups is hard.
- Manual analysis is inefficient.

3. PROBLEM ROOT CAUSE — RC

- Data is stored in unprocessed formats and lacks visualization tools.

6. PROBLEM ROOT CAUSE — BE

- Data is stored in unprocessed formats and lacks visualization tools.
- Decision-making is slow and fragmented.

4. TRIGGERS — TR

- When tasked to analyze heart disease data, stakeholders manually review spreadsheets and reports.

5. EMOTIONS: BEFORE / AFTER — EM

- **Before:** Overwhelmed, slow, uncertain.
- **After:** Confident, fast, and informed.

7. YOUR SOLUTION — SL

- Developed interactive Tableau dashboard with:
  - Demographic filters (age, gender, region)
  - Lifestyle indicators (smoking, BMI, cholesterrol)
  - Risk factor comparison charts
  - Regional heatmaps,
  - Storyteling dashboards,
  - "Use as Filter" features

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9. OUTCOME — O

- Solution effectively addresses identified problems.
- Enhanced efficiency and decision-making speed with interactive visual analysis.